

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - 7

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Class : SYIT	Batch : 1
Date of Assignment : 05/08/26	Date/Time of Submission : 05/08/26

a. Implement a hash table with separate chaining for collision handling.

code:

```
size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)].append((key,value))

insert(10, "apple")
insert(15, "banana")
insert(20, "mango")
insert(32, "guava")
insert(33, "pineapple")

print(table)
print("iqra S017")
```

output:

```
== RESTART: C:/Users/mvluc/AppData/Local/Programs/Python/Python314/tyertyrt.py =
[[ (10, 'apple'), (15, 'banana'), (20, 'mango') ], [], [(32, 'guava')], [(33, 'pineapple')], []]
iqra S017
|
```

b. Store and retrieve data from the hash table.

CODE:

```
size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)].append((key,value))

def search(key):
    for k,v in table[key%size]:

        if k==key:
            return v
    return "NOT FOUND"

insert(10, "apple")
insert(15, "banana")
insert(20, "mango")
insert(32, "guava")
insert(33, "pineapple")

print(table)
key=int(input("Enter Key"))
print(search(key))

print("iqra S017")
```

OUTPUT:

```
===== RESTART: C:/Users/mvluc/AppData/Local/Programs/Python/Python31
[[ (10, 'apple'), (15, 'banana'), (20, 'mango') ], [], [(32, 'guava')], [(33, 'pineapple')], []]
Enter Key 20
mango
iqra S017
|
```

NOT FOUND

```
===== RESTART: C:/Users/mvluc/AppData/Local/Programs/Python/Python31
[[ (10, 'apple'), (15, 'banana'), (20, 'mango') ], [], [(32, 'guava')], [(33, 'pineapple')], []]
Enter Key 25
NOT FOUND
iqra S017
```

